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Dose-response effects of microaeration on thermophilic lignocellulose-based anaerobic digestion: performance, stability and microbial community dynamics

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ABSTRACT

Microaeration has emerged as a promising strategy for enhancing anaerobic digestion (AD) of lignocellulosic biomass, although inconsistent results highlight the need for further study. This study tested three different aeration rates (1, 5 and 8 mLO_2/gVS) on anaerobic co-digestion of wheat straw and cattle manure under thermophilic conditions (55 °C) in continuous-stirred tank reactors with working volume of 60 L. Daily injection of 5 mLO_2/gVS achieved the highest methane yield, with a 20.6% increase compared to the non-aerated control digester. Under these conditions, cellulose and hemicellulose degradation reached maximum levels of 75% and 66.3%, respectively. In the microaerated digester, remarkable changes in microbial population were observed. Injection of 1 mLO_2/gVS and 5 mLO_2/gVS increased the relative abundance of *Lentimicrobium*, *Turicibacter*, and *unclassified MBA03* as well as hydrogenotrophic *Methanothermobacter* which could contribute to enhanced lignocellulose degradation and biogas generation. At the final and highest oxygen dose (8 mLO_2/gVS), methane yield began to gradually decrease. Under these conditions, the microbial profile shifted back to that of the non-aerated digester, suggesting that the activity of certain microbial groups was inhibited. These findings highlight the potential of controlled microaeration to enhance both methane yield and lignocellulosic degradation at larger scales, offering valuable insights for optimizing AD processes.

1. Introduction

The annual generation of agricultural biomass amounts to 5.5 billion tons, and due to its high availability, it is an attractive and reliable feedstock for anaerobic digestion (AD) (Archana et al., 2024). In recent years, agricultural biogas production has emerged as a favorable strategy for generating alternative energy while reducing greenhouse gas (GHG) emissions (Møller et al., 2022). Consequently, it has a positive impact on the climate and environment, while AD process also yields valuable digestate, which supports nutrient recycling (Jasińska et al., 2023; Vaneeckhaute et al., 2013). However, using agricultural waste as a substrate for AD poses several challenges. Agricultural biomass is often rich in lignocellulose, whose complex and recalcitrant structure hinders efficient biogas production. Lignocellulosic biomass consists of polysaccharides like cellulose and hemicellulose embedded in a lignin matrix, forming a highly complex three-dimensional structure that impedes the hydrolysis step during AD (Manyi-Loh et al., 2013). The crystallinity of cellulose and the complexity of lignin (as an aromatic polymer) pose another barrier hindering the biodegradation efficiency in AD systems (Millati et al., 2020).

Pre-treatment is often applied before subjecting lignocellulosic feedstock to AD. To date, various pre-treatment methods have been explored to enhance lignocellulose-based AD, including alkaline or acid treatment, steam explosion, and ozonation (Karthikeyan et al., 2024), as

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well as hyper-thermophilic AD (Kozera et al., 2024). Although these pretreatment techniques can yield good results, they have specific drawbacks, such as high energy and/or chemical consumption, inhibitory effects, and challenges in scaling up the process. These limitations can hinder commercial biogas production due to negative economic returns or adverse environmental impacts due to the release of inhibitory by-products (Xu et al., 2021).

In recent years, microaeration has gained increased interest as an attractive strategy for enhancing the AD process. Microaeration, defined as the introduction of small quantities of oxygen or air into an AD system, has been reported as a promising technology with a positive impact on overall methane yield (Zeb et al., 2022), and stabilization of the process (Nguyen et al., 2019). Additionally, microaerobic conditions play a key role in hydrogen sulfide (H_2S) removal during AD (Andreides et al., 2021). Microaerobic conditions can also alter the diversity and abundance of microbial communities, particularly hydrolytic bacteria (Fu et al., 2016) leading to increased enzymatic activity of extracellular hydrolytic enzymes (e.g., cellulases, xylanases, lipases). This enhancement improves the hydrolysis of complex materials such as lignocellulose (Duarte et al., 2018; Nguyen and Khanal, 2018). In that sense, applying microaeration can facilitate the AD process by creating a unique environment that supports the coexistence of facultative microbes and obligate anaerobes. In this setup, facultative microorganisms consume trace amounts of oxygen for hydrolytic processes, while preserving anaerobic conditions for obligate anaerobic microbes (Fu et al., 2023).

Despite its potential, some studies reported no improvement or even adverse effects of the microaeration on AD performance (Díaz et al., 2011a; Fdz.-Polanco et al., 2009). In a study by Tsapekos et al. (2021), low and high oxygen loads (5 and 15 mLO_2/gVS) were single-injected into the working volume of the reactor on daily basis for 35 days and resulted in positive (+10%) and negative (−4%) impact on batch AD tests, respectively. However, the injection of 7.3 mLO_2/gVS improved the methane yield by 7% on continuous mode. In the continuous mode, the microaeration was done after the adaptation process lasting 15 days and was performed utilizing a single injection of oxygen into the working volume of the continuous-stirred tank reactor (CSTR) every day for 30 days. These results vary because of different aeration strategies as well as environmental and operating conditions in AD.

In most cases, microaeration of AD has been investigated at laboratory-scale (Amin et al., 2021; Cao et al., 2022; Zhu et al., 2022), and only a few studies reported the application of microaeration on an industrial-scale but mainly with a focus on H_2S removal from a headspace of the reactors (Huertas et al., 2020; Jeníček et al., 2017; Muller et al., 2024). In that sense, microaeration technology has not yet been fully explored and conflicting results cannot indicate the exact underlying mechanism for an improved AD process. Hence, the investigation of the process on a larger scale may potentially help in its better understanding since it is still under development (Chen et al., 2020); (Krayzelova et al., 2015a) and might fill some knowledge gaps in terms of affecting factors or operational conditions. Furthermore, the microaeration treatment on larger scale has been done so far mostly by subjecting liquid substrates to the AD system such as high-solid wastewater (Duangmanee et al., 2007), primary sludge (Nghiem et al., 2014), and wastewater ((Niu et al., 2016); (Pokorna-Krayzelova et al., 2018)). Thus, the utilization of agricultural wastes i.e. lignocellulosic materials in the large scale microaeration is attractive as it can extend the knowledge about the process on its impacts on solid substrates as well.

The diversity in microaeration implementation along with many knowledge gaps in the literature calls for further investigation to shed light on lignocellulose-based AD processes. To the best of authors' knowledge, there are only few studies reporting the impact of microaeration on lignocellulose-based AD which are limited to small laboratory-scale. Due to the potential of microaeration to enhance the hydrolysis rate with no adverse effect on the AD process, more investigation in this field is crucial to expand and promote its application on a larger scale. Thus, the present work tested three different microaeration conditions in an anaerobic digester with a 60 L working volume, fed with lignocellulosic feedstock, and provided new insights into AD process dynamics and microbial community changes. This study hypothesized that controlled microaeration can enhance hydrolysis efficiency and stabilize the AD process without compromising biogas production. The primary objectives were to evaluate the impact of different microaeration strategies on process performance, assess microbial community shifts, and identify optimal conditions for scaling up lignocellulose-based AD.

2. Materials and methods

2.1. Feedstock and inoculum

Wheat straw (WS) was harvested locally in Oostkamp, Belgium, and shredded to particle size <1 cm with a cutter mill. Cattle manure (CM) was collected from a dairy farm in Adinkerke, Belgium, and stored at 4 °C before further use. Both shredded wheat straw (SWS) and CM were then used as feedstock for the AD experiments. The inoculum was sourced from a thermophilic agro-industrial biogas installation in Pittem, Belgium. The characteristics of the feedstock and the inoculum are given in Table 1.

2.2. Experimental design and set-up

2.2.1. Digesters operational conditions

The anaerobic co-digestion experiments were conducted under thermophilic conditions (54 ± 1 °C) using two single-stage continuous stirred-tank reactors (CSTR) with working volume of 60 L. One of the digesters was utilized to perform micro-aeration treatment whereas the second one was used as the control (referred later in the paper as MA and CON, respectively). Thermophilic conditions were selected due to faster organic breakdown, especially complex materials e.g. agricultural wastes, than under mesophilic conditions (Labatut et al., 2014). The hydraulic retention time (HRT) was 30 days and the organic loading rate (OLR) for each reactor was 2.70 ± 0.10 gVS/L/day. Initially, the OLR was 3.40 gVS/L/d but due to operational issues observed in the beginning and to avoid overfeeding, it was decreased. The mixing ratio of SWS and CM was 40:60 (VS basis) to provide more balanced nutrients and buffering capacity throughout the process (Krishania et al., 2013; Z. Song and Zhang, 2015). CM was selected as a co-substrate due to diverse microbial communities, which can readily adjust to varying operational circumstances. In addition, CM was noted as a material containing readily degradable organic matter that boosts microbial activity, facilitating disruption of lignocellulose recalcitrance of wheat straw (Kadam et al., 2023). Hence, CM and SWS was co-digested to provide improved process performance leading to higher, and more stable methane production through synergy.

Table 1
Physical-chemical characteristics of feedstock and inoculum. CM: cattle manure, SWS: shredded wheat straw.

Parameter	CM	SWS	Inoculum
pH [−]	7.46 ± 0.12	8.01 ± 0.18	8.21 ± 0.19
TS [%]	6.67 ± 0.47	89.61 ± 0.35	3.79 ± 0.18
VS [%TS]	75.44 ± 1.32	89.74 ± 0.02	72.16 ± 0.61
Cellulose content [%TS]	32.69 ± 0.23	42.59 ± 0.97	–
Hemicellulose content[%TS]	17.73 ± 1.50	29.66 ± 0.82	–
Lignin content [%TS]	8.22 ± 0.31	5.56 ± 0.42	–
TKN [g/L]	4.28 ± 0.36	4.26 ± 0.64	–
TAN [g/L]	2.40 ± 0.40	n.d.	–
VFA [mgCOD/L]	1469 ± 12	n.d.	–

Where TS: total solids, VS: volatile solids, TKN: total Kjeldahl nitrogen, TAN; total ammonium nitrogen, VFA: volatile fatty acids, and n.d.: not detected. Note: mean value ± standard deviation from triplicate measurements.

Continuous mixing of the reactors was provided by a central shaft with a propeller at the center and the bottom, rotating at 60 rpm. The experiment ran for 92 days and was divided into two phases. The first phase is called the adaptation period involving the first 21 days of start-up in which no aeration was applied (Period 0); followed by the lowest dose of air injection between days 22–44 (Period 1). The second phase involves the injection of higher air doses (Period 2 and Period 3) into the AD system between days 45–66 and 67–92, respectively, and referred as the experimental phase of the study. The adaptation period was done to acclimatize the microbial community to microaeration and the real influence of microaeration was evaluated by the results obtained in the experimental phase (Periods 2–3). The details of the microaeration are described in Section 2.2.2.

Biogas production was daily recorded in both reactors employing a Ritter Drum-Type Gas Meter TG0.5/0.4 mbar (Ritter Apparatebau GmbH & Co. KG, Bochum, Germany) with measuring range of 1–60 L/h. The gas composition was measured every day by utilizing a portable BIOGAS 5000 analyzer (Chelmsford, UK). The analyzer was connected to the reactor gas outlet to measure methane and carbon dioxide concentrations, and a single measurement lasted at least 40 s. Gas composition was normalized to 100%, assuming that only significant components in the dry biogas are methane and carbon dioxide. To monitor the stability of the AD process, samples of digestate from each reactor were collected once a week for pH, total solids (TS), volatile solids (VS), volatile fatty acids (VFA), and ammonium-nitrogen (NH_4^+-N) content.

2.2.2. *Microaeration treatment*

To provide an efficient microaeration treatment, one of the CSTR's was re-designed and equipped with a U-shape metal pipe with three perforations (diameter ~ 1 mm) at the bottom. The illustrative scheme of the micro-aerated reactor (MA) is shown in Fig. 1. The air pump was connected to the flow meter to supply accurate portion of air through the inlet port. Then the air was distributed into the reactor through the U-shape pipe. The pipe was attached to the inlet port and sealed at the end to provide equal air distribution inside the reactor. The end of the pipe was also attached to the side of the biogas outlet not to tamper with the headspace of the reactor which might affect the biogas composition. The air was injected through the inlet port and released at the bottom of the pipe through perforations.

Three different doses of oxygen (1, 5 and 8 mLO_2/gVS per day) were tested. The choice of doses was inspired by the investigations of Zeb et al. (2022) and Tsapekos et al. (2017). They reported the aeration dose of around 5 mLO_2/gVS as the rate with the highest enhancement in biogas production, and 10 mLO_2/gVS as a rate leading to instability of AD process at lab-scale. The air was injected into the reactor once a day after feeding using air pump Tom&Co Aqua Bubble 33 (Tom& Co, London, UK). The air injection was controlled by a gas flowmeter SHLLJ LZQ-7 (Yuyao Shunhuan Flowmeter Co., Zhejiang, China) which was steadily set up at 1 L/min. The oxygen dosing strategy was calculated based on predetermined aeration rates for each experimental period.

Table 2
Microaeration dosing strategy: daily oxygen demands and corresponding air delivery parameters.

Period	Daily VS input [gVS/day]	Aeration rate [mLO_2/gVS]	Demanded oxygen portion [mLO_2]	Demand air portion [mL]	Contact time [sec]
Period 1	191	1	191	909	55
Period 2	191	5	955	4548	273
Period 3	191	8	1528	7276	437

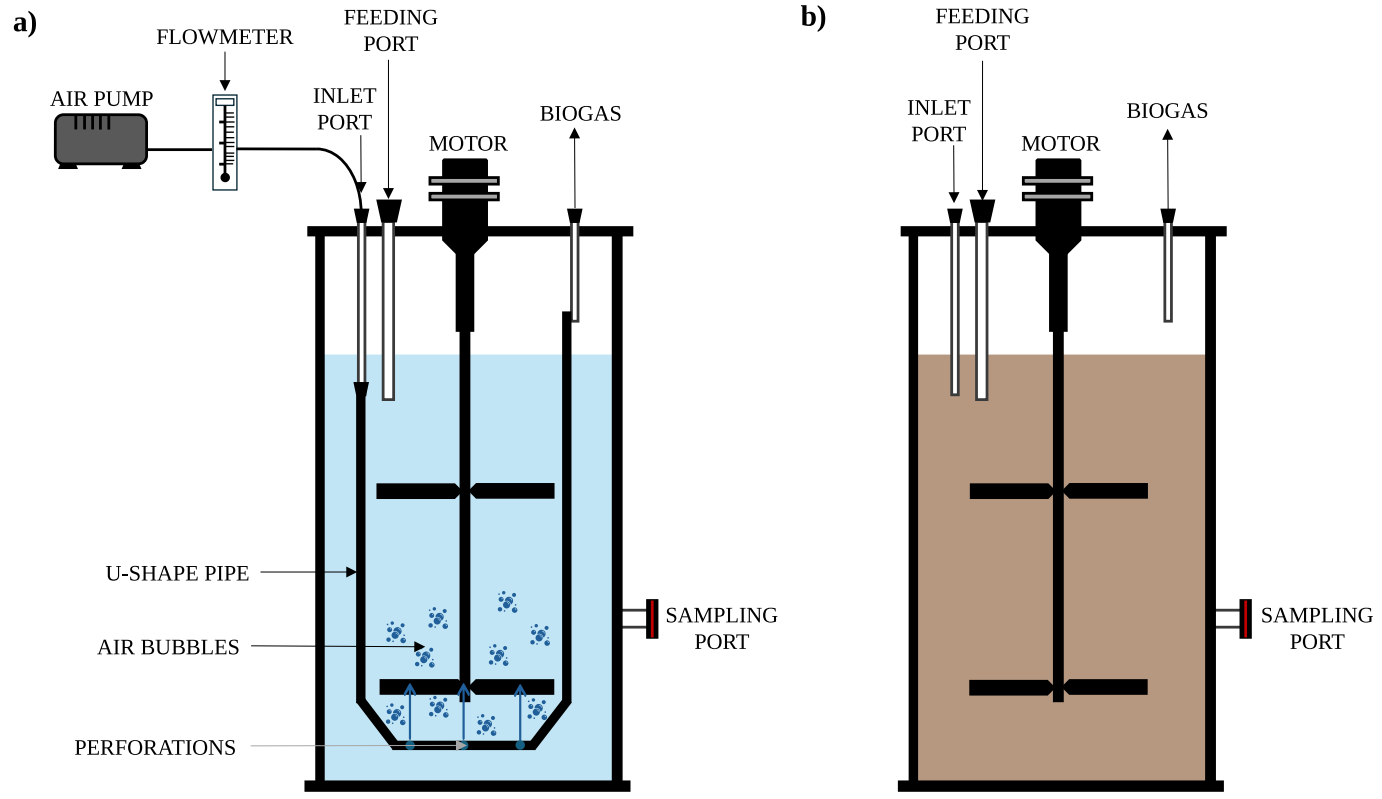


Fig. 1. Schematic illustration of the continuous stirred-tank reactor (CSTR) configurations: (a) micro-aerated digester with integrated oxygen distribution system via perforated U-shape pipe and (b) conventional non-aerated control digester.

Aeration rates and calculated dosing parameters for each experimental period have been depicted in Table 2.

Daily oxygen demand was calculated by multiplying the daily VS input (191 g) by the corresponding aeration rate, yielding oxygen requirements of 191, 955, and 1528 mLO_2 per day, respectively, for three consecutive periods. Since atmospheric air (with approximately 21% oxygen content) was injected rather than pure oxygen, the required air volume was calculated by dividing the daily oxygen demand by 0.21, resulting in daily air doses of 909, 4548, and 7276 mL/day. To supply the target volumes using the air pump equipped with a flowmeter set to 1 L/min, the pump operating times were computed as 55, 273, and 437 s per day for Period 1, 2, and 3, respectively (Eq. (1)).

$$Contact\ time\ \left[\frac{sec}{day}\right] = \frac{Demanded\ air\ portion\ \left[\frac{mL}{day}\right] \cdot 60\ sec}{1000\ mL} \quad (1)$$

At the end of each period, digestate samples from both reactors were collected to measure pH, TS, VS, VFA, NH_4^+-N content and analyze fiber degradation and microbial community changes.

2.3. Analytical methods

The pH was measured using pH meter Eutech PCD 6500 (Thermo Fisher Scientific Inc., Waltham, MA, USA). The TS was determined as the residual after placing the pre-weighed samples of feedstocks and digestates for 24 h drying in hot-air oven at 105 °C (Aronsson et al., 2022). The VS was determined as the difference between the dry weight and the inorganic content, obtained by placing the dry weight in a muffle oven at 550 °C for 4 h (APHA, 2012). The TKN was measured via the Kjeldahl method via destruction, steam distillation, and titration employing Foss Kjeltec™ 8400 (Foss, Hillerød, Denmark). The ammonium-nitrogen (NH4 + -N) was determined by steam distillation employing Gerhardt Vapodest 20 (C. Gerhardt GmbH & Co.KG, Königswinter, Germany) and titrimetric method utilizing 718 Stat Titrino (Metholm AG, Herisau, Switzerland) (InnoLab, 2021a). VFAs concentrations were analyzed by gas chromatography (GC) with a flame ionization detector-GC-FID Trace 1300 GC (Thermo Fisher Scientific Inc., Waltham, MA, USA) (InnoLab, 2021b). The VFA concentrations were calculated as chemical oxygen demand (COD) equivalents. The COD equivalents were adopted from elsewhere (Atasoy et al., 2020) and were 1.067 gCOD /g acid for acetic acid, 1.514 gCOD /g acid for propionic acid, 1.818 gCOD /g acid for isobutyric and butyric acids, 2.039 gCOD /g acid for isovaleric and valeric acids, 2.207 gCOD /g acid for caproic acid. The analyses were done in triplicate for feedstocks and inoculum characteristics, however the samples of digestates taken once a week for process monitoring were analyzed in a single replicate. Fiber analysis was performed using the filter bag system under pressurization employing ANKOM ^{DELTA} Fiber Analyzer (ANKOM® Technology, Macedon, NY, USA). The neutral detergent fiber (NDF) as well as acid detergent fiber (ADF) analysis was done following the operator's manual. Acid-detergent lignin (ADL) was done manually by soaking the bags in 72% H_2SO_4 for 3 h as reported by Van Soest et al., 1991. Additionally, the bags were ash-corrected following the method found in Silva et al., 2018. Each sample was analyzed in triplicate, and the fiber analysis was done sequentially.

2.4. Microbial community analysis

Samples from the sludge used as inoculum, and from day 21, day 44, day 66 and 92 in both CON and MA digesters were extracted using PowerSoilPro (QiaGen) with a beat-beating step of 5 × 4000 rpm for 15″ with 45″ intervals using a PowerLyzer instrument (QiaGen). The DNA was eluted in 100 μL elution buffer. 10 μL genomic DNA extract was sent out to LGC genomics GmbH (Berlin, Germany) for library preparation and sequencing on an Illumina Miseq platform with v3 chemistry with the primers 341F (5'-CCT ACG GGN GGC WGC AG -3') and 785Rmod (5'-GAC TAC HVG GGT ATC TAA KCC-3') (Klindworth et al., 2013).

The DADA2 R package was used to process the amplicon sequence data according to the pipeline tutorial (Callahan et al., 2016). In the first quality control step, the primer sequences were removed, and reads were truncated at a quality score cut-off (truncQ = 2). Besides trimming, additional filtering was performed to eliminate reads containing any ambiguous base calls or reads with high expected errors (maxEE = 2,2). After dereplication, unique reads were further denoised using the Divisive Amplicon Denoising Algorithm (DADA) error estimation algorithm and the selfConsist sample inference algorithm (with the option pooling = TRUE). The obtained error rates were inspected and after approval, the denoised reads were merged.

Finally, the ASV table obtained after chimera removal was used for taxonomy assignment using the Naive Bayesian Classifier and the DADA2 formatted Silva v138.1 (Quast et al., 2013). Alpha diversity of the 16S rRNA marker gene was calculated using the inverse Simpson index, beta diversity was calculated using Bray–Curtis distances and PCoA was used to make ordination plots. The relative abundances of the top bacteria and archaea genera were selected to study specific dynamics in the most abundant genera (Beterams et al., 2022).

2.5. Mass balance

A mass balance was performed on each reactor to determine the efficacy of each applied aeration rate in fiber biodegradation during the whole micro aeration treatment. The efficacy of lignocellulose decomposition refers to the degradation levels which were calculated as a relative value measuring the difference between the initial and final concentration of a specific component as a percentage (Eq. (2)) and were calculated for TS, VS and main lignocellulose constituents i.e. fraction of hemicellulose, cellulose, and lignin.

$$Degradation\ level\ (\%) = \frac{C_{inf} - C_{eff}}{C_{inf}} \cdot 100 \quad (2)$$

2.6. Statistics

The average values and standard deviations were calculated for key monitored parameters, including methane and carbon dioxide relative concentrations, hydrogen sulfide concentrations, biogas and methane yields, and fiber constituents' content. To define statistically significant differences ($p < 0.05$) among experimental measurements, a two-sided student *t*-test for two independent groups was done utilizing MS Excel calculation program. In addition, principal component analysis (PCA) was done for each experiment period to indicate dose-relationship effects of microaeration on the process. The PCA analysis was performed using R program.

3. Results

3.1. Impact of microaeration on biogas production and methane yield

Daily biogas and methane yields of the MA and CON digesters are shown in Fig. 2a and b, respectively, and the calculated biogas and methane yields are given in Table 3. Methane generation in the first 44 days (Period 0 + Period 1) fluctuated in both digesters. In the MA reactor, the biogas yield varied from 230 to 614 L/kgVS per day and the methane yield fluctuated from 112 to 302 L/kgVS per day. Similar values were found in the CON digester, in which the daily biogas yields varied from 227 to 545 L/kgVS and the methane yields ranged between 117 and 313 L/kgVS. In this period, no significant difference in biogas and methane production was observed ($p > 0.05$). The shifts in biogas and methane production in this adaptation period could be caused by changes in operational conditions at the beginning of the experiment during which the OLR value lowered from 3.40 to 2.70 gVS/L per day.

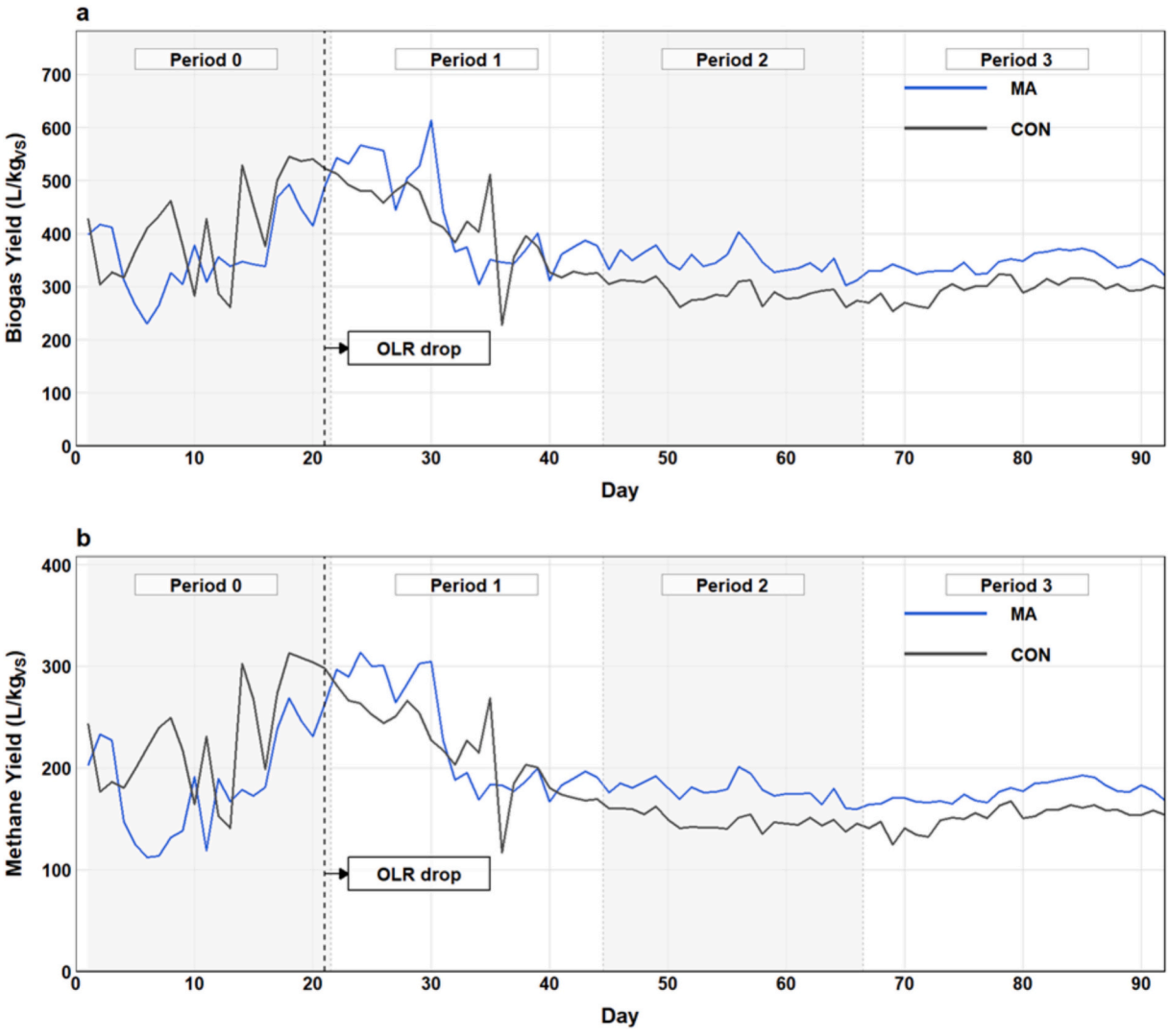


Fig. 2. Daily biogas (a) and methane (b) yields in the MA and the CON reactor during the experiment. The dashed lines separate the periods of the experiment during which different aeration conditions were applied (Period 0 = no aeration, Period 1 = 1 mLO_2/gVS, Period 2 = 5 mLO_2/gVS, Period 3 = 8 mLO_2/gVS).

Table 3
Biogas and methane production in micro-aerated (MA) and non-aerated (CON) digesters across three experiment periods.

	Period 1		Improvement**	Period 2		Improvement**	Period 3		Improvement**
Aeration rate [mLO_2/gVS]	1			5			8		
Digester ID	MA	CON		MA	CON		MA	CON	
Biogas yield [L/kgVS]	**429** ±93	**405** ±79	- 5.93%	**347** ±23	**288** ±18	+20.49%*	**344** ±16	**296** ±18	+16.22%*
Methane yield [L/kgVS]	**228** ±54	**215** ±43	- 6.05%	**178** ±11	**147** ±8	+21.09%*	**176** ±9	**153** ±10	+15.03%*

Note: mean value ± standard deviation value from daily measurements ($n = 23$ for Period 1, $n = 22$ for Period 2, $n = 26$ for Period 3), * statistically different with p-value threshold <0.05, ** calculated as relative difference between results from microaerated and non-aerated rector for each experimental period and expressed as percentage, N/A: not assessed.

Lowering the OLR value was necessary to provide an appropriate stability of the process. Initially, under high OLR conditions during Period 0, the FOS/TAC parameter varied between 0.50 and 0.73 and 0.47–0.54 $mgHAc/mgCaCO_3$ in the MA and the CON reactor, respectively. During Period 1, these values lowered and ranged from 0.29 to 0.50, and 0.23–0.44, respectively, presenting still relatively high fluctuations. The FOS/TAC parameter reached a stable level after lowering the OLR value to 2.70 gVS/L per day during Period 2 and Period 3 and ranged between 0.20 and 0.26 $mgHAc/mgCaCO_3$ (Table S1a, Table S1b).

The actual impact of microaeration on biogas and methane

generation can be noticed during Periods 2 and 3, where steady-state conditions were reached in both reactors and daily biogas generation rates were at relatively stable levels. The highest biogas and methane yields were found in the MA reactor during Period 2 with injection of 5 mLO_2/gVS, where the yields averaged 347 ± 23, and 178 ± 11 L/kgVS, respectively. At the same time, the biogas production in the CON reactor was 288 ± 18 L/kgVS and methane yield equaled 147 ± 8 L/kgVS, which demonstrate significant differences in biogas and methane production ($p < 0.05$) during Period 2. Thus, the injection of 5 mLO_2/gVS led to the enhancement of the biogas production by 20% and the methane yield by 21%.

A similar pattern was obtained in the final part of the experiment where 8 mLO_2/gVS was injected into the MA digester in Period 3. Significant differences in biogas and methane yields were noted ($p < 0.05$). The average biogas yield obtained in the period amounted to 344 ± 16 L/kgVS in the MA reactor and 296 ± 18 L/kgVS in the CON reactor providing the enhancement of biogas production by 16%. In the case of methane generation, the yields were 176 ± 9 L/kgVS and 153 ± 10 L/

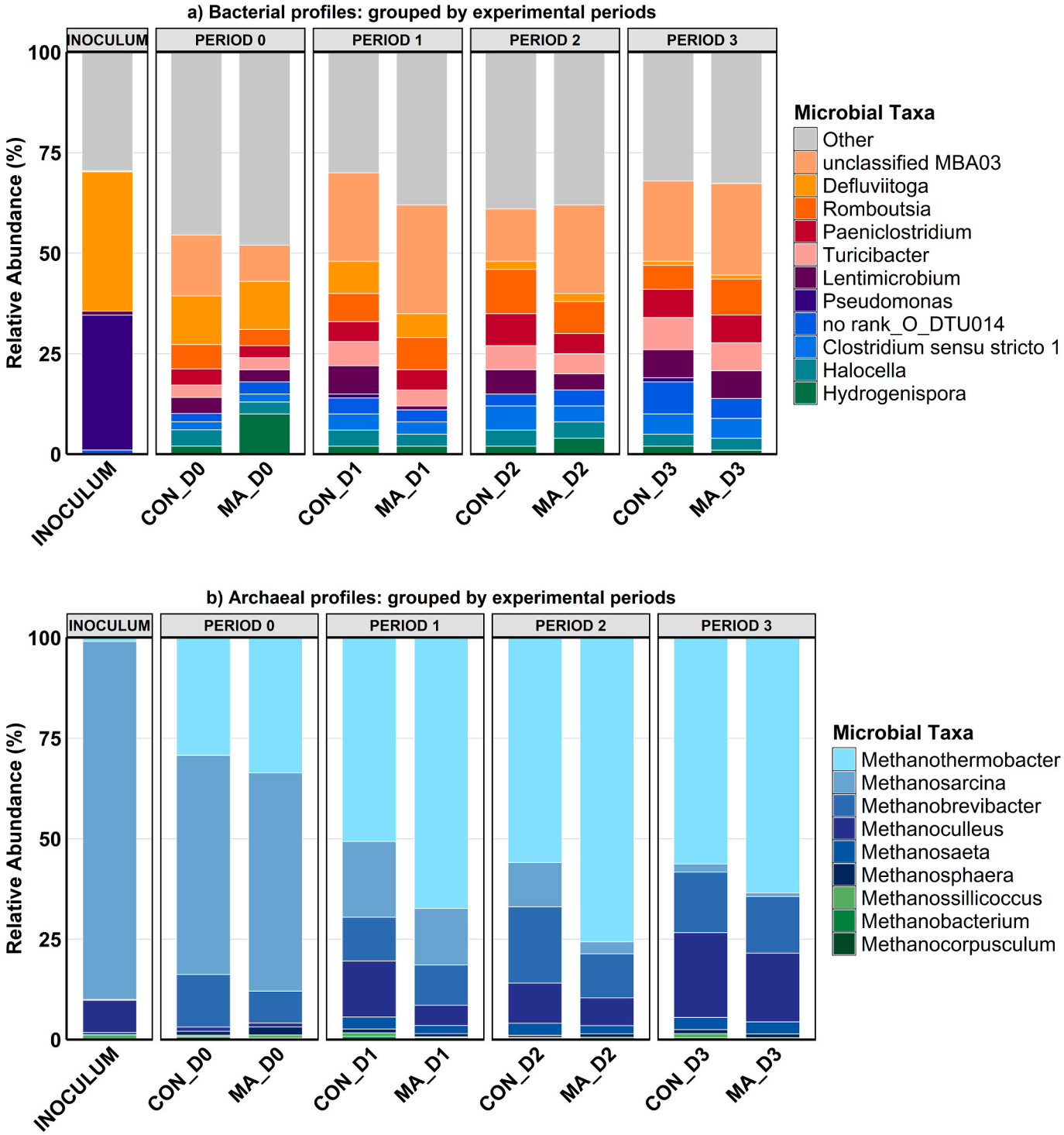


Fig. 3. Relative abundance of (a) top bacterial genera and (b) and methanogenic archaea as amplicon sequence variants (ASVs). Top genera were calculated based on their combined relative abundance across all samples and plotted as ASVs, all other genera were labeled as "Other".

kgVS, respectively in the MA and CON digesters which led to increased methane production by 15%. This means that the injection of 8 mLO_2/gVS into the MA digester resulted in enhanced biogas and methane production but with slightly lower efficiency when compared to the results noted during Period 2. The lower efficacy of methane production could be potentially caused by changes in microbial communities' profiles which is discussed in Section 3.4. Overall, our results highlighted the aeration rate of 5 mLO_2/gVS as the most promising microaeration strategy in terms of methane production, and the aeration rate of 8 mLO_2/gVS already seemed like inhibiting the methanogenic activity (see Fig. 3).

3.2. Impact of microaeration on digestion stability

The stability of the microaeration treatment was assessed by comparing the selected process parameters in both reactors as given in Table 4. Microaeration treatment did not influence the overall conventional AD process parameters like TS, VS, pH, and TAN. TS and VS content remained stable throughout the whole experiment and equaled 5.89 ± 0.13%, 5.96 ± 0.16%, and 75.5 ± 0.94%, 74.6 ± 1.20% TS for the MA and the CON reactors respectively. The pH value in the MA reactor varied from 8.08 to 8.22 and was similar to the range of pH observed in the CON reactor (8.06–8.21). In the case of TAN concentrations, it was noted that the concentration slightly increased in both digesters. However, they also remained at relatively constant levels and varied from 2.15 to 2.35 g/L throughout the whole experimental period. This indicates that microaeration had no adverse effects on AD and can result in the process stability akin to the conventional digestion process. Conversely, notable observations were made regarding the concentrations of total VFAs. The total VFA concentrations in the MA digester were 1594, 754, and 494 mgCOD/L for Period 1, Period 2 and Period 3, respectively, whereas in the CON digester these values were 785, 1036, and 461 mgCOD/L, respectively.

Changes in biogas composition under different aeration rates are given in Table 4. Even though the relative methane and carbon dioxide contents were similar and ranged between 51 and 53%, the remarkable differences were observed in H_2S concentrations. The lowest H_2S concentrations were found in the MA reactor after the injection of 5 and 8 mLO_2/gVS which were 2092 ± 123, and 2234 ± 268 ppm, respectively. These values were significantly different ($p < 0.05$) compared to the concentration analyzed in the CON reactor with an average 2477 ± 203, and 2555 ± 138 ppm, respectively. To understand better the impact of microaeration on H_2S mitigation, we evaluated the H_2S concentration fluctuations inside the MA reactor over time. The monitoring of biogas composition before, during, and after the injection of air was done and can be found in the Fig. S1. Accordingly, H_2S concentration before air injection was over 3000 ppm in the MA reactor and remarkably decreased right after air injection to the level of around 500 ppm. In the following 5 h, H_2S concentration decreased down to 318 ppm with the final H_2S removal up to 89%. However, after 24 h, H_2S concentration reached again the level of 2851 ppm. These results suggest that even a single injection of oxygen into the reactor can lead to gradual H_2S

Table 4
Performance comparison of micro-aerated (MA) and non-aerated (CON) digesters across three experimental periods.

	Period 1		Improvement**	Period 2		Improvement**	Period 3		Improvement**
Aeration rate [mLO_2/gVS/d]	1			5			8		
Digester ID	MA	CON		MA	CON		MA	CON	
Process performance									
TS [%]	**5.27**[a] ±0.59	**5.74**[a] ±0.92	N/A	**6.12**[a] ±0.52	**6.14**[a] ±0.66	N/A	**5.98**[a] ±0.22	**6.22**[a] ±0.71	N/A
VS [%TS]	**74.83**[a] ±1.41	**75.62**[a] ±1.55	N/A	**76.56**[a] ±2.57	**75.52**[a] ±0.64	N/A	**75.88**[a] ±1.32	**75.84**[a] ±0.88	N/A
pH [-]	**8.11**[a] ±0.07	**8.14**[a] ±0.05	N/A	**8.14**[a] ±0.07	**8.09**[a] ±0.06	N/A	**8.12**[a] ±0.09	**8.21**[a] ±0.10	N/A
TAN [g/L]	**2.05**[a] ±0.11	**2.07**[a] ±0.09	N/A	**2.22**[a] ±0.10	**2.30**[a] ±0.08	N/A	**2.33**[a] ±0.07	**2.37**[a] ±0.06	N/A
Total VFAs [mgCOD/L]	**1594**[b]	**785**[b]	N/A	**1032**[a]	**1062**[a]	N/A	**614**[a]	**725**[a]	N/A
Acetic acid	**1081**[b]	**58**[b]		**685**[a] ±153	**732**[a] ±159		**425**[a] ±69	**505**[a] ±143	
Propionic acid	**327**[b]	**83**[b]		**208**[a] ±115	**207**[a] ±57		**58**[a] ±23	**89**[a] ±47	
Isobutanoic acid	**31**[b]	**15**[b]		**18**[a] ±2	**18**[a] ±4		**17**[a] ±2	**17**[a] ±1	
Butanoic acid	**18**[b]	**11**[b]		**14**[a] ±1	**12**[a] ±1		**13**[a] ±2	**13**[a] ±0	
Isovaleric acid	**53**[b]	**20**[b]		**27**[a] ±7	**25**[a] ±5		**23**[a] ±2	**23**[a] ±1	
Valeric acid	**27**[b]	**22**[b]		**25**[a] ±3	**24**[a] ±1		**25**[a] ±2	**25**[a] ±1	
Caproic acid	**57**[b]	**46**[b]		**54**[a] ±6	**44**[a] ±14		**54**[a] ±5	**53**[a] ±2	
Gas composition									
CH_4 concentration [%]	**52.90**[c] ±2.57	**53.00**[c] ±1.07	N/A	**51.20**[c] ±1.02	**51.20**[c] ±1.33	N/A	**51.20**[c] ±0.81	**51.50**[c] ±1.09	N/A
CO_2 concentration [%]	**47.10**[c] ±2.57	**43.00**[c] ±1.07	N/A	**48.80**[c] ±1.02	**48.80**[c] ±1.33	N/A	**48.80**[c] ±0.81	**48.50**[c] ±1.09	N/A
H_2S concentration [ppm]	**3359**[c] ±866	**3044**[c] ±429	-10.35%	**2092**[c] ±123	**2477**[c] ±203	+15.54%*	**2234**[c] ±268	**2555**[c] ±138	+12.56%*
Fiber composition									
Hemicellulose content [g/kg]	**9.39**[d] ±0.14	**9.75**[d] ±0.28	-3.69%	**7.37**[d] ±0.32	**9.28**[d] ±0.36	+20.58%*	**7.74**[d] ±0.28	**9.18**[d] ±0.11	+15.69%*
Cellulose content [g/kg]	**10.64**[d] ±0.30	**11.98**[d] ±0.17	+11.19%*	**8.53**[d] ±0.39	**12.07**[d] ±0.38	+29.33%*	**8.73**[d] ±0.43	**12.07**[d] ±0.10	+27.67%*
Lignin content [g/kg]	**6.66**[d] ±0.25	**6.64**[d] ±0.21	-0.30%	**6.58**[d] ±0.24	**6.77**[d] ±0.27	-2.81%	**6.80**[d] ±0.14	**6.64**[d] ±0.11	-2.41%

Note: [a] mean value ± standard deviation value of weekly monitoring measurements ($n = 4$ for Period 1, $n = 3$ for Period 2, n = 4 for Period 3), [b] due to high fluctuations of VFAs in both reactors, values from single measurement (without standard deviation value presented) recorded on the last monitoring day for Period 1 were depicted, [c] mean value ± standard deviation value from daily measurements (n = 23 for Period 1, n = 22 for Period 2, n = 26 for Period 3), [d] mean value ± standard deviation value from quadruplicate measurements, * statistically different with p-value threshold <0.05, ** calculated as relative difference between results from microaerated and non-aerated reactor for each experimental period and expressed as percentage, N/A: not analyzed.

removal from the biogas. Factors that might be responsible for the phenomenon are discussed in the Discussion section.

The contents of lignocellulosic compounds such as cellulose, hemicellulose, and lignin found in digestates collected from the CON and the MA reactor at the end of each applied aeration rate is given in Table 4. The degradation levels of these compounds were calculated following Eq. (2). Contents of cellulose, hemicellulose, and lignin in effluent (C_{eff}) were 34.08 ± 0.14, 21.85 ± 2.47, and 6.99 ± 0.14, respectively. The lignocellulosic compounds found in the MA reactor were comparable with the CON reactor during the adaptation period (Period 1) where the aeration rate of 1 mLO_2/gVS had no influence on lignocellulose degradation. During this period, cellulose was degraded by 68.78 ± 1.22%, and 64.84 ± 1.83% in the MA and the CON reactor, respectively. Similarly, the hemicellulose degradation levels were found as 57.02 ± 3.81%, and 55.41 ± 3.39%, respectively.

During the experimental phase of the experiment (Period 2 and Period 3), the prominent influence of microaeration on lignocellulose degradation was observed. The highest cellulose and hemicellulose degradation levels were determined in Period 2 when 5 mLO_2/gVS was injected into the MA reactor. The application of the dose resulted in significant differences ($p < 0.05$) in the contents of these saccharides reaching degradation rates of 74.97 ± 0.58 and 66.26 ± 2.10%, respectively. It enhanced the degradation of cellulose by 16% and hemicellulose by 15%, compared to the CON digester which were 64.59 ± 1.27%, and 57.54 ± 2.86%, respectively.

Comparable findings were noted for Period 3 with the aeration rate of 8 mLO_2/gVS. During Period 3, cellulose and hemicellulose in the MA reactor degradation levels were 74.38 ± 0.50%, and 64.60 ± 2.44%, respectively. In the CON reactor, these values equaled 64.57 ± 2.04%, and 58.00 ± 3.82%, which resulted in 15% and 11% higher cellulose and hemicellulose breakdown, respectively. In the case of lignin content, no significant differences were observed during the whole experiment ($p > 0.05$). In the first 44 days, lignin was degraded only by 4.64 ± 2.27%, and 5.00 ± 2.69% in the MA and the CON reactors, respectively. Lignin content did not change significantly in both reactors and its degradation levels varied from 2.74 to 5.82% in the following periods, suggesting that applied aeration rates had negligible influence on lignin decomposition. Taken together, the results indicate that the injection of 5 mLO_2/gVS into the AD system provides favorable conditions to enhance the biodegradability of lignocellulosic feedstock.

3.3. Impact of microaeration on microbial communities

Microbial population analysis is a crucial factor in assessing microaeration treatment's potential for AD. Starting with an inoculum predominantly composed of the phyla Pseudomonadota (ex-Proteobacteria) (34%) and Thermotogota (34%), both digesters selected for Bacillota (ex-Firmicutes) among bacteria, increasing their proportion from 21% in the inoculum to 63% and 61% for CON and MA respectively (Period 0). Similarly, the proportion of archaea increased to 13 and 14% from an initial 3% in the inoculum. This shift, observed after 21 days (Period 0), can be attributed to the anaerobic conditions that drastically reduced the proportion of the aerobic *Pseudomonas* (only representative of the Proteobacteria) to 0.2 and 0.4% in CON and MA, respectively. Thermotogota were affected to a lesser extent, remaining at 13% in both digesters. These changes were linked to an increase in overall microbial diversity, as reflected in the Inverse Simpson index, which rose from 7.67 in the inoculum to 22.3 and 28.2 in CON and MA, respectively (Table S2). This increased diversity is primarily due to Bacillota growth, dominated by *unclassified MBA03* by the end of Period 1 (15 and 9% in CON and MA, respectively). *Unclassified MBA03* has been identified recently as key player in anaerobic digesters (Otto et al., 2024; Puchol-Royo et al., 2023). The increased diversity was also due to the development of archaea and especially of *Methanosarcina*, *Methanothermobacter* and, to a lesser extent, *Methanobrevibacter*. These methanogens, essential in AD, predominantly follow the hydrogenotrophic pathway, with *Methanosarcina* also capable of acetoclastic methanogenesis (CONRAD, 2020).

Beta diversity analysis highlighted the growing dissimilarity of the two populations over time. The Principal Component Analysis (PCA) showed the dominant effect of time on the bacterial community in both CON and MA digesters (Fig. S2), with three major clusters emerging of this analysis: i) Period 0 samples, ii) Periods 1 and 2 samples, and iii) Period 3 samples. These results indicate that temporal factors outweigh microaeration pressure in driving community differences.

The analysis of the population dynamics (see Fig. 3) showed a consistent selection of *unclassified MBA03* (approximately 20% in both digesters by the end of Period 3) and stable proportions of fermentative genera from the Clostridia class, including *Clostridium* sensu stricto, *O_DTU014, Paeniclostridium* and *Romboutsia*. *Lentimicrobium* and *Turicibacter* were also notable, exceeding 7% of total genera at the end of Period 3. The roles and implications of these genera are discussed in Section 4. In addition to the general trend of microbial community evolution over time, the injection of oxygen during Period 1 (1 mLO_2/gVS) had a significant effect on specific genera. *Unclassified MBA03* exhibited a 200% increase in MA, compared to a 50% increase in CON. This increase can be associated with the representation of the hydrogenotrophic *Methanothermobacter*, which more than doubled in MA (8%) compared to CON (3%) following Period 1. Further increases in oxygen flow rate to 5 mLO_2/gVS by the end of Period 2 reinforced this trend, with significantly higher proportions of *unclassified MBA03* (22 versus 13%) and *Methanothermobacter* (10 versus 6%) in MA compared to CON. Thus, certain microbial groups were promoted under microaeration conditions at 5 mLO_2/gVS during Period 2, leading to enhanced methane yield and lignocellulose degradation. In contrast, the microbial community profile in the aerated digester during Period 3 resembled that of the non-aerated digester. This suggests that the activity of certain microbial genera may have been inhibited under these conditions, indicating that oxygen tolerance during microaeration narrowed at doses up to 8 mLO_2/gVS.

Among all the genera identified, no significant presence of known sulfur-reducing bacteria (SRB) was detected throughout the experiment (<1%). However, *Defluviitoga*, a putative SRB (Hania et al., 2012), was the main representative of the Thermotogota phylum. Its proportion decreased over time in both CON and MA reactors, dropping from 7% and 6% in Period 1 to 2% in Period 2 and further to 1% in Period 3, respectively. This decline suggests that microbial production of H_2S occurred in both digesters and explains the significant reduction in H_2S levels between Periods 1 and 2/3 (Table 3).

The microaeration conditions likely inhibited *Defluviitoga* activity, contributing to the observed rapid decrease in H_2S concentrations in the MA reactor (Fig. S1), as shown by Gutierrez et al., 2008. However, this inhibition does not appear to be the primary factor driving the decrease in *Defluviitoga* abundance, as a similar trend was observed in the CON reactor. Additionally, no notable abundance of sulfur-oxidizing bacteria (SOB) was detected (<1%), indicating that no significant microbial mitigation of H_2S occurred in the MA reactor compared to the CON reactor.

3.4. Dose-response effects of microaeration on the AD process

The Principal Component Analysis (PCA) was done to assess dose-response relationships between process performance and microbial community kinetics across three microaeration conditions (Fig. 4). The results revealed distinct patterns of microbial-performance links that evolved with increasing doses of oxygen. During Period 1, the confidence ellipses for MA and CON clusters depicted substantial overlap that indicates no remarkable differentiation at this low oxygen dose. This corresponds to no significant differences in methane yield and fiber degradation. On the other hand, in Period 2, the most pronounced separation between MA and CON samples was observed. The MA_D2 samples clustered in the positive region of PC1, strongly associated with

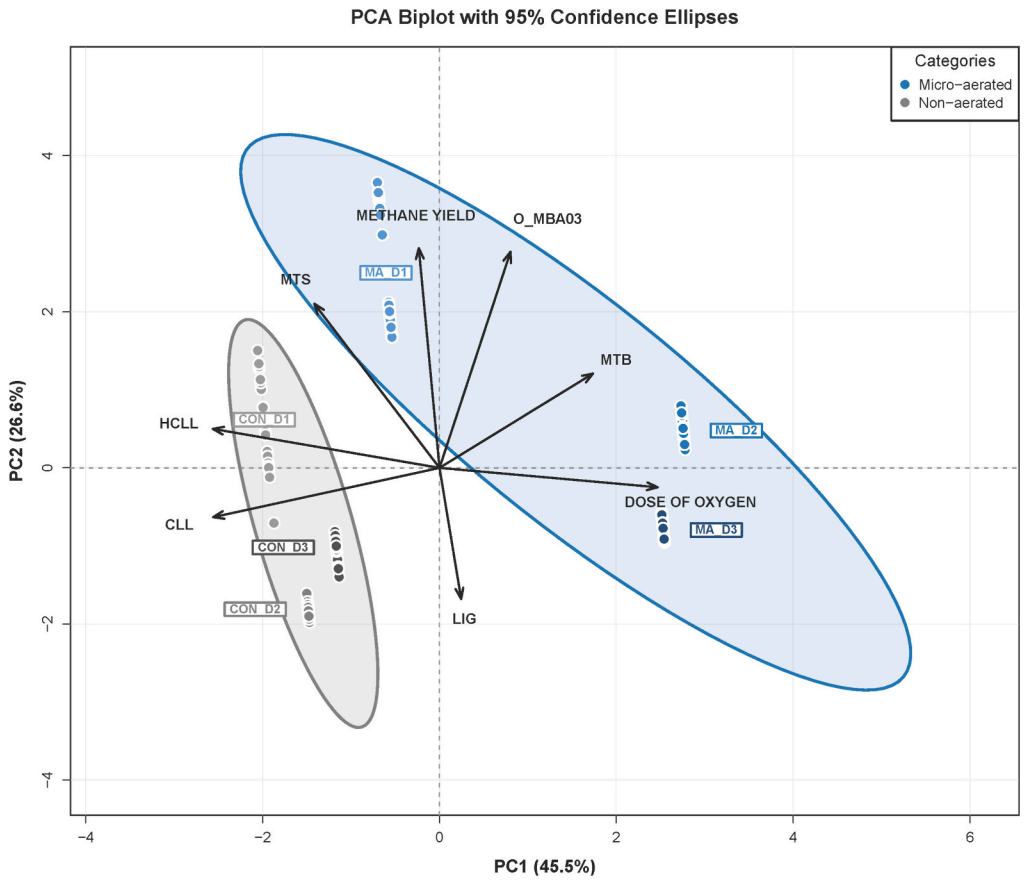


Fig. 4. Principal component analysis (PCA) of dose-dependent microaeration effects on the anaerobic digestion performance and microbial community kinetics. **Note:** CLL: cellulose fraction content, HCLL: hemicellulose fraction content, LIG: lignin fraction content, MTS: *Methanosarcina* abundance, MTB: *Methanothermobacter* abundance, O_MBA03: *unclassified MBA03* abundance. Variable arrows were scaled by factor of 5 for better visualization.

the methane yield vector and key microbial genera. This clear separation refers to the significant improvement in process performance during this period. The methane yield increased by 20.6% in microaerated digester when compared to non-aerated digester. Also, cellulose degradation reached 75.0% in MA samples compared to 64.6% in CON samples, and hemicellulose degradation level amounted to 66.3% versus 57.5% (Table 4). The positioning of CLL and HCLL vectors on the opposite side from MA samples confirms lower fiber content, which characterizes higher degradation under microaerated conditions. Moreover, for Period 2, microbial community analysis proved enhanced abundance of unclassified MBA03 (22% in MA samples versus 13% in CON samples) and hydrogenotrophic *Methanothermobacter* (10% vs. 6%). It explains the strong association of these genera with MA samples on the PCA biplot. In Period 3,the PCA results reflect a decreased enhancement of microaeration when compared to control. The separation of MA and CON samples for this Period is still remarkable but smaller than during Period 2. The methane yield improvement decreased to 15%, and microbial communities' profiles became similar between the digesters (Fig. 3). This differentiation might indicate that dose of 8 mLO_2/gVS may be inhibitory for oxygen-sensitive populations participating in syntrophic acetate oxidation and hydrogenotrophic methanogenesis.

In general, the PCA results clearly demonstrate that microaeration under optimal dosing strategy with 5 mLO_2/gVS creates distinct microbial and functional differences between MA and CON digesters. It was mostly driven by the boost of syntrophic acetate-oxidizing bacteria (unclassified MBA03) and hydrogenotrophic *Methanothermobacter*, which facilitated improved lignocellulose degradation and methane generation. The clear spatial separation of MA and CON samples along with the strong association between methane yield and key microbial genera, gives quantitative evidence that 5 mLO_2/gVS demonstrates the optimal microaeration rate for thermophilic anaerobic co-digestion of LCB.

4. Discussion

This study investigated three different microaeration rates (1, 5, and 8 mLO_2/gVS) to assess the feasibility of microaerobic treatment on large lab-scale for enhanced methane production and lignocellulose hydrolysis. Each aeration rate was evaluated for approximately one HRT. According to Holliger et al. (2016), stable digestion can be assumed when measured parameters (e.g. biogas production, VFA, VS, pH) are consistently maintained within 10% of their average values for a minimum period of one HRT. In our study, the results demonstrate stable process performance with consistent key parameters (Table 4). It confirms the system had reached operational stability within each evaluation period.

The results revealed that the injection of 1 mLO_2/gVS into the reactor during the adaptation period did not influence the AD performance and ran similarly to the conventional AD process. This finding conflicts with some previous reports, for example, (Zhu et al., 2022) reported an enhancement of 7.8% in methane yield with a single injection of oxygen load of 0.2 mLO_2/gVS while performing AD of corn straw. However, the difference might be related to the scale of the experiments. The enhancement was obtained under small lab-scale conditions utilizing shaking air bath with the shaking speed of 200 rpm. That could potentially provide better operational conditions for oxygen distribution inside the reactor and enhanced the overall performance of the AD when compared to larger digesters. This indicates the necessity of

investigating the impact of microaeration on AD on a larger scale, suggesting that the reactor volume may potentially influence the accessibility of oxygen inside the reactor.

Even though the aeration rate of 1 mLO_2/gVS resulted in no significant impact on enhanced methane yield, remarkable differences in methane generation were observed during the following periods. The highest methane yield enhancement was obtained with an oxygen load of 5 mLO_2/gVS and amounted to 20%. A slightly lower increase (16%) was noted when 8 mLO_2/gVS was employed. These results tie well with previous studies wherein the methane yield was increased by 7.2%, injecting the load of oxygen of 5 mLO_2/gVS for consecutive 3-day treatment during the AD of wheat straw (Tsapekos et al., 2017). A similar conclusion was reached by Fu et al. (2015) who reported that microaerobic pretreatment of corn straw with the same oxygen load led to an enhancement of methane yield by 16% when compared to untreated digester. However, in both cases, microaeration was done at small lab-scale conditions as well and deployed as a pre-treatment of substrate instead of direct injection inside the digester.

In another study, Tsapekos et al. (2021) conducted microaeration experiment in which wheat straw was co-digested with livestock slurry in CSTR and reported that a load of oxygen 7.3 mLO_2/gVS resulted in the highest enhancement of methane yield by 7%. The enhancement was caused by the promotion of hydrolysis during microaeration treatment but no remarkable changes in microbial community were observed. Only an increased relative abundance of aero-tolerant archaea e.g. *Methanosarcina* and *Methanobacterium* under microaerobic condition was observed which can suggest that these species contributed to increase in methane generation. The same microaeration strategy was used by Zeb et al. (2022) and observed greatly higher methane yield enhancement up to 32% during the AD of buffalo manure under mesophilic conditions on a lab scale. In that case study it was reported to degrade acid-soluble lignin (ASL) content by 78% which could consequently enhance the availability of enzymes to decompose cellulose and hemicellulose that resulted in higher conversion of the substrate into biogas. However, in both papers, authors do not indicate a particular reason for enhancement in obtained methane yields. Either way, these results suggest that microaerobic conditions affect the methane yield positively but at different levels, depending on the dose of oxygen, operational conditions (e.g., scale, reactor configuration, frequency of microaeration), and choice of feedstock.

Nevertheless, our study concludes that microaeration at a flow rate of up to 8 mLO_2/gVS positively enhanced methane yield from the AD of wheat straw and cattle manure. It also highlights that oxygen dosages used in laboratory settings can have practical applications and lead to significant differences in methane production on a larger scale. At 8 mLO_2/gVS/d, the microbial community shifted but remained largely similar to that of the non-aerated digester, suggesting that oxygen doses above this threshold may begin to inhibit certain microbial activity. The depletion of microbial activity could explain the gradual decrease in methane yield observed during Period 3. However, while 8 mLO_2/gVS may still be a promising flow rate for lignocellulose-based AD, further investigation is required.

Another important effect of the microaeration on AD process is a change in H_2S concentrations which is believed to be mitigated in situ by providing microaerobic conditions inside the digester (Krayzelova et al., 2015b). For instance, it was observed that microaeration provided enhanced methane yield with simultaneous H_2S mitigation from AD of chicken manure which generates biogas with a high concentration of H_2S (Mahdy et al., 2020; Y. Song et al., 2020). In our study, application of 5 and 8 mLO_2/gVS resulted in significant differences ($p < 0.05$) in H_2S concentration between the MA and the CON reactors. Considering the average values of H_2S concentration in the reactors, its content decreased by 16% and 9% in the MA during D2 and D3 compared to the untreated digester, respectively. According to Y. Song et al. (2020), a single injection of oxygen at 1.4 mL/gVS/d resulted in 68% H_2S removal and was improved after the implication of continuous mode of oxygen injection reaching the removal level of 31–60%. It was also observed that continuous aeration with an oxygen load of 4.2 mL/gVS/d led to a removal level in the range of 91–120%. In the study of Mahdy et al. (2020), it was reported that H_2S content decreased by ~50% after a single injection of 6.3 mLO_2/gVS/d. Thus, it indicates that in situ desulfurization might be also performed when subjecting cattle manure and wheat straw to co-digestion but with a lower efficiency.

It is noteworthy that 5 h after the injection of air, the H_2S was removed at the level of ~90%. However, after 24 h, its concentration reached the same concentration value as it was before the application of oxygen. Even though, the long exposure to a single injection of oxygen could contribute to the gradual mitigation of H_2S in the biogas composition, the continuous injection of air might be necessary for its long-term consistent removal and should be investigated in further research. H_2S removal found in the MA reactor might be also explained by the suggestion found in Song et al., 2020 which stated that enhancing hydrogenotrophic methanogenesis with air injection could help maintain lower hydrogen partial pressure and promote AD stability. Hypothetically, changes in H_2S concentrations could be also attributed to the competitive oxygen utilization by sulfur oxidizing bacteria (SOB) (Nghiem et al., 2014) or by the inhibition of SRB while oxygen injection inside the digester. Some SRB species can oxidize sulfide under aerobic conditions and reduce sulfate under oxic conditions (Canfield and Des Marais, 1991; Fuseler, 1996). However, there is no clear evidence so far indicating the growth of SRB under these conditions. On the other hand, in our study, among all the genera identified, no significant presence of SRB or SOB were detected which makes this hypothesis doubtful. Alternatively, the phenomenon can be related to the limited solubility of pure oxygen in water. The injected air could consequently transfer through the liquid fraction and interact with rich H_2S biogas in the headspace of the digester. Then, the temporary changes in H_2S concentrations could have been mitigated due to the biochemical or chemical oxidation process (Díaz et al., 2011b; Ramos et al., 2012) Summing up, the exact metabolic pathway of sulfide conversion and oxidation under microaerobic conditions is still poorly understood (Krayzelova et al., 2015a) and it is challenging to indicate only one underlying mechanism responsible for H_2S removal under microaeration treatment.

The degradation level of lignin fraction was constant in both reactors throughout the experiment, and no remarkable differences have been observed since lignin is believed to be a very recalcitrant material with a low level of biodegradability when subjected to the AD. Lignin decomposition poses a very challenging task due to its complex chemical structure which is resistant to the AD process and might require additional treatment that will be combined with microaeration strategy. For example, Akyol et al. (2019) revealed that fungal pre-treatment of lignocellulosic biomass with *Trametes versicolor* improved methane yield by 10–18% and cellulose degradation up to 80%. In the mentioned study, fungal treatment and aerobic conditions provided favorable conditions for lignin decomposition leading to increased presence of specific hydrolytic enzymes and consequently resulting in enhanced lignocellulosic breakdown. It suggests that combined treatment should be also taken into consideration in future.

Microaerobic conditions have been reported to facilitate the abundance of hydrolytic bacteria (Fu et al., 2016) whose enzymatical activities might lead to improved hydrolysis of complex biomass such as lignocellulose (Duarte et al., 2018; Nguyen and Khanal, 2018). An example of the functional consortia that can hydrolyze polysaccharides in lignocellulose i.e. hemicellulose and cellulose are the phyla Bacillota and Bacteroidetes (Hanreich et al., 2013). The relative abundance of both phyla has been also reported to increase under microaerobic conditions (Fu et al., 2016; Zhu et al., 2022) creating a unique microbial population permitting coexistence of facultative microbes and obligate anaerobic microorganisms (Fu et al., 2023). In our study, the enhancement of relative abundance of phyla Bacillota was observed (see Fig. 3) which potentially contributed to increased degradation levels of

hemicellulose and cellulose in the MA reactor after injection of 5 and 8 mLO_2/gVS. The hemicellulose degradation in the microaerated reactor was enhanced by 8.72% and 6.60% during Period 2 and Period 3, respectively, whereas the cellulose fraction decomposition was enhanced by 10.38% and 9.82%, respectively.

The genera identified in the final sample (after Period 3) are commonly found in anaerobic digesters (sources) and appear to play key roles in the main functions of AD. Among these, *Lentimicrobium* and *Turicibacter* are responsible for starch and fat degradation, respectively. The various genera of *Clostridia* present are likely involved in oxidizing resulting molecules into acetate (and CO_2) while producing H_2. Subsequently, *unclassified MBA03*, identified as a syntrophic acetate-oxidizing bacterium (SAOB) (Dyksma et al., 2020), completes the oxidation of acetate and other volatile fatty acids (VFA) into CO_2, making it available for hydrogenotrophic methanogens.

This analysis suggests that improving AD performance could involve enrichment strategies focused on genera such as *Lentimicrobium* or *Turicibacter*, which are significantly represented and likely responsible for degrading complex organic compounds. Enhancing their activity could optimize the utilization of lignocellulose-based substrates.

5. Conclusion

Microaeration of a lignocellulose-based anaerobic digester at a single-injection rate of 5 mLO_2/gVS resulted in the highest methane yield and the greatest degradation of hemicellulose and cellulose. These conditions increased the relative abundance of hydrogenotrophic methanogens (*Methanothermobacter) and SAOB (unclassified MBA03)*, which may have facilitated a shift from acetoclastic methanogenesis to hydrogenotrophic methanogenesis due to syntrophic acetate oxidation. This shift, in turn, contributed to in situ H_2S removal. Additionally, the results demonstrated that microaeration at 5 mLO_2/gVS significantly influenced the microbial community, promoting the growth of *Lentimicrobium*, *Turicibacter*.Based on these findings, controlled microaerobic conditions can be practically applied to lignocellulose-based AD at a larger scale without adverse effects on the process. However, the authors suggest further investigation considering a continuous dosing of oxygen prior to an upscaled implementation.

CRediT authorship contribution statement

Marcin P. Kozera: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Çağrı Akyol:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Florent Bouchon:** Writing – original draft, Investigation, Formal analysis. **Ramon Ganigué:** Writing – review & editing, Supervision. **Bernard Willems:** Resources, Methodology. **Erik Meers:** Supervision, Resources, Conceptualization. **Kai Bester:** Writing – review & editing, Supervision. **Henrik B. Møller:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biteb.2026.102593.

Data availability

Data will be made available on request.

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